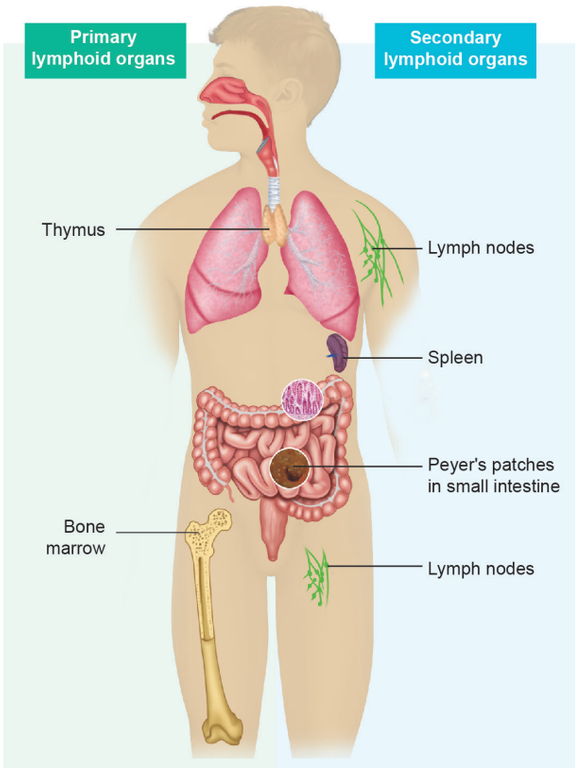
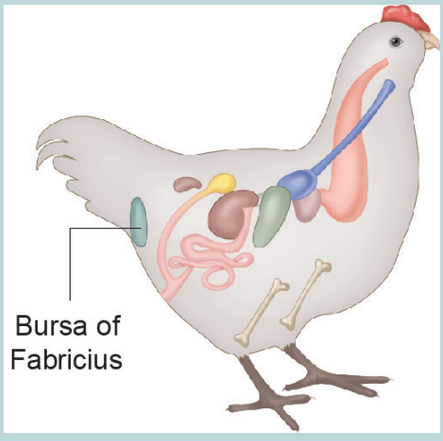


Fig. 8.3 Lymphoid organs in human body

Based on their functions, they are classified into **primary or central lymphoid organs** and **secondary or peripheral lymphoid organs**. The primary lymphoid organs provide appropriate environment for lymphocytic maturation. The secondary lymphoid organs trap antigens and make it available for mature lymphocytes, which can effectively fight against these antigens.



Bursa of Fabricius is a primary lymphoid organ of birds. It is attached to the dorsal side of the cloaca. B lymphocytes mature in the bursa and bring about humoral immunity.

Primary lymphoid organs

Bursa of Fabricius of birds, bone marrow and thymus gland of mammals constitute the primary lymphoid organs involved in the production and early selection of lymphocytes. These lymphocytes become dedicated to a particular **antigenic specificity**. Only when the lymphocytes mature in the primary lymphoidal organs, they become **immunocompetent cells**. In mammals, B cell maturation occurs in the bone marrow and T cells maturation occurs in the thymus.

Thymus

The thymus is a flat and bilobed organ located behind the sternun, above the heart. Each lobe of the thymus contains numerous lobules, separated from each other by connective tissue called septa. Each lobule is differentiated into two compartments, the outer compartment or **outer cortex,** is densely packed with immature T cells called thymocytes, whereas the inner compartment or medulla is sparsely populated with mature thymocytes. One of its main secretions is the hormone **thymosin**. It stimulates the T cell to become mature and **immunocompetent**. By the early teens, the thymus begins to atrophy and is replaced by adipose tissue **(Fig. 8.4).** Thus thymus is **most active** during the **neonatal and pre-adolescent periods**.

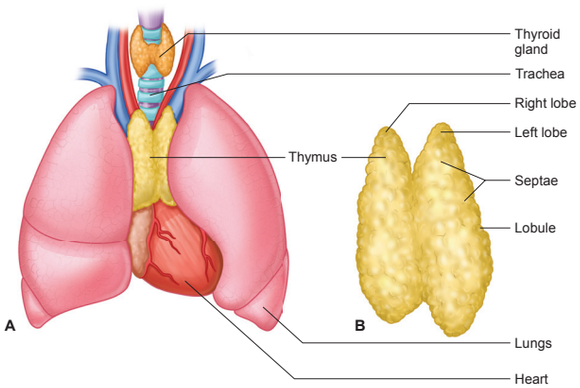


Fig. 8.4 Primary lymphoid organ - Thymus A) Location B) Structure

XII Std Zoology Chapter 8 EM.indd 133 23/12/2022 16:43:14